How Model Structure and Parameter Choices Influence Tumor Immune Dynamics and Treatment Response

AMATH 881
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1 Introduction

From the first report in 1904 of tumor regression during viral infections to various immune cells being discovered and clinical interventions being approved; the role and interactions of the immune system in regards to cancer has been well researched thoroughly scientific methods [2], revealing complex interactions. This paper builds on the model by Hernandez-Lopez et al. [4] inspired by Elewaut et al. [4] and supplemented by Reis et al. [7] further extending by exploring the parameter choices described in the article and extending the model to include macrophage and chemotherapy interactions; allowing for further biological interpretation of the complex system. In Section 2, the original model from [4] is revisited, highlighting incompatible parameter choices with interpretation along side investigation of vague parameter definition. Section 3 explores a macrophage extension; identifying the tumor-free equilibrium and characterizing its stability. A chemotherapy extension is presented in Section 4, also contained exploration of the tumor-free equilibrium point.

2 Original Model

Below is the model introduced in Hernandez-Lopez et al. [4] with $C(t)$ representing the density of cancer cells, $T(t)$ the density of T-Lymphocytes cells, $M_n(t)$ the density of monocytes and $M(t)$ the density of macrophages. This work was inspired by the experiments preformed in Elewaut et al. [1] which highlighted the importance of the MCHI cross dressing between monocytes and T-cells captured by the term βMT.

$$\begin{aligned}
\frac{dC}{dt} &= rC(1-bC)(C-\gamma) - \alpha CT, \\
\frac{dT}{dt} &= \lambda C + \beta MT - \delta T, \\
\frac{dM}{dt} &= \rho - \eta M - \omega CM.
\end{aligned}$$

The experiments suggested that by the MCHI cross dressing mechanism, monocytes can activate

the stimulation of T-cells. Further research from DeNardo et al. [3] suggest that monocytes can also suppress the stimulation of T-cells thus allowing the parameter β to take on negative values. This has an effect on the tumor free equilibrium state for this model as stated in [4]: the tumor-free equilibrium $E_0 = (0, 0, \rho/\eta)$. The stability of this equilibrium is as follows.

(a) E_0 is locally asymptotically stable if $\beta\rho < \delta\eta$.

(b) E_0 is a saddle if $\beta\rho > \delta\eta$.

(c) E_0 is a degenerate equilibrium if $\beta\rho = \delta\eta$.

Thus contributing to the following proposition:

Proposition 1. *E_0 is locally asymptotically stable when $\beta < 0$ and $\rho, \eta, \delta > 0$*

Proof. Using proposition 2.1 from [4], when $\beta < 0 \implies \beta\rho < \delta\eta$ for strictly positive values of ρ, η, δ. □

Based on the literature from [3], this seems to be a biologically plausible state, yet no dynamics of negative β values where explored in the original paper. Another note on the model, is that within the paper, it was stated that η can be set to zero, justifying that the $-\omega CM$ term would capture the apoptosis mechanism captured by the $-\eta M$ term due to the dominant nature of cancer mediated monocyte death. Yet, upon further inspection, setting $\eta = 0$ causes $\rho = 0$ thus biologically invalidating the model as this implied that monocytes have no production mechanism.

3 Macrophage Extension

The system for macrophage extension as follows:

$$\begin{aligned}
\frac{dC}{dt} &= rC(1 - bC)(C - \gamma) - \alpha CT - aCM_c \\
\frac{dT}{dt} &= \lambda C + \beta M_n T - \delta T + \phi T M_c \\
\frac{dM_n}{dt} &= \rho - \eta M_n - \omega C M_n - g M_n M_c \\
\frac{dM_c}{dt} &= -d M_c M_n - e T M_c + h
\end{aligned}$$

This is the system for macrophage extension. $C(t)$ denotes the density of cancer cells, $T(t)$ the density of T-Lymphocytes cells, $M_n(t)$ the density of monocytes and $M_c(t)$ the density of macrophages. The base of the model comes form Hernandez-Lopez, as described in section 2. The additional term in the cancer cell equation of $-aCM_c$ captures the inhibitory effects of macrophages on the cancer cells [7]. $\phi T M_c$ represents the stimulatory effect of macrophages on T Lymphocytes. Whereas, the term $-gM_nM_c$ models the maturation of monocytes into macrophages. The additional equation $\frac{dM_c}{dt}$ aims to capture simplified dynamics of the macrophage interactions within the tumor microenvironment with $-dM_cM_n$ representing the reduction in monocytes after expansion of inflammatory monocytes as noted in Elewaut et al., h refers to a general recruitment rate for macrophages and

$-eTM_c$ capturing the observed absence of macrophages in tumor areas with T cells absent as well [1].

Proposition 2. *The point* $P_0 = \left(0, 0, M_n^* = \frac{d\rho - gh}{dn}, M_c^* = \frac{hn}{d\rho - gh}\right)$ *is the tumor free equilibrium of the system with stability as follows:*

1. P_0 *is asymptotically stable if* $\beta M_n^* + \phi M_c^* < \delta$
2. P_0 *is saddle if* $\beta M_n^* + \phi M_c^* > \delta$
3. P_0 *is degenerate if* $\beta M_n^* + \phi M_c^* = \delta$

Proof. To analyze the stability of P_0 it is necessary to compute the Jacobian of the system:

$$J = \begin{pmatrix} -r(\gamma + C(-2+3bC-2b\gamma)) - \alpha T - aM_c & -\alpha C & 0 & -aC \\ \lambda & \beta M_n - \delta + \phi M_c & \beta T & \phi T \\ -\omega M_n & 0 & -\eta - \omega C - gM_c & -gM_n \\ 0 & -eM_c & -dM_c & -dM_n - eT \end{pmatrix}$$

Substituting the tumor free equilibrium gives:

$$\begin{pmatrix} -r\gamma - aM_c^* & 0 & 0 & 0 \\ \lambda & \beta M_n^* - \delta + \phi M_c^* & 0 & 0 \\ -\omega M_n^* & 0 & -n - gM_c^* & -gM_n^* \\ 0 & -eM_c^* & -dM_c^* & -dM_n^* \end{pmatrix}$$

The eigenvalues of this matrix are:

$$\begin{aligned} \xi_1 &= -r\gamma - aM_c^* \\ \xi_2 &= \beta M_n^* - \delta + \phi M_c^* \\ \xi_{3,4} &= \frac{-(n + gM_c^* + dM_n^*) \pm \sqrt{(n + gM_c^* + dM_n^*)^2 - 4ndM_n^*}}{2} \end{aligned}$$

For stability, all eigenvalues must have negative real parts. For $\xi_{1,3,4}$ this is true for all values of the parameter, leaving ξ_2 to be the predictor of stability in the system. □

4 Chemotherapy Extension

Below is the model which further extends the model from Hernandez-Lopez [4] by introducing Chemotherapy terms alongside the macrophage extension discussed in section 3.

$$
\begin{aligned}
\frac{dC}{dt} &= rC(1-bC)(C-\gamma) - \alpha CT - \kappa CD - aCM_c \\
\frac{dT}{dt} &= \lambda C + \beta M_n T - \delta T + \phi T M_c - qTD \\
\frac{dM_n}{dt} &= p - \eta M_n - \omega C M_n - g M_n M_c \\
\frac{dM_c}{dt} &= -dM_c M_n - eTM_c + h - iM_c D \\
\frac{dD}{dt} &= u - \tau D
\end{aligned}
$$

$C(t)$ denotes the density of cancer cells, $T(t)$ the density of T-Lymphocytes cells, $M_n(t)$ the density of monocytes, $M_c(t)$ the density of macrophages and $D(t)$ represents the drug concentration within the tumor microenvironment over time. $\frac{dD}{dt}$ follows a standard model for drug concentration with the parameter u capturing the peak drug concentration and $-\tau D$ modeling exponential decay. Additional terms in the cancer, T-lymphocytes and macrophage equations were added to capture the effect of chemotherapy reducing healthy cells in addition to cancer cells [8]. The monocyte equation does not have an additional term based on the assumption chemotherapy does not have an effect on monocyte production.

To assess the tumor free equilibrium it is necessary to split this paper moves forward with the assumption that the chemotherapy steady state also reaches eradication. This is biologically plausible as it represents the withdraw of chemotherapy after cancer eradication. With this assumption two cases unfold. The first is no T-cells are present within the tumor microenvironment, the second represents an active immune system in the TME after cancer eradication corresponding to $T^* > 0$. Both cases are explored as follows:

Proposition 3. *The cancer, T-cell and drug free equilibrium point* $P_0 = (0, 0, \frac{dp-hg}{d\eta}, \frac{h\eta}{dp-hg}, 0)$ *has stability characterization as follow:*

1. P_0 *is asymptotically stable if* $\beta M_n^* + \phi M_c^* < \delta$
2. P_0 *is saddle if* $\beta M_n^* + \phi M_c^* > \delta$
3. P_0 *is degenerate if* $\beta M_n^* + \phi M_c^* = \delta$

Proof. Finding the cancer and drug free equilibrium is found by setting $C^*, D^*, u, T^* = 0$

$$
\begin{aligned}
\frac{dM_n}{dt} &= p - M_n(\eta + gM_c) = 0 \implies M_n^* = \frac{p}{\eta + gM_c^*} \\
\frac{dM_c}{dt} &= h - M_c(dM_n) = 0 \implies M_c^* = \frac{h}{dM_n^*}
\end{aligned}
$$

The Jacobian of the chemotherapy system is:

$$\begin{pmatrix} -r(\gamma+C(-2+3bC-2b\gamma))-aM_c-\kappa D & -aC & 0 & 0 & -\kappa C \\ \lambda & \beta M-\delta+\phi M_c-qD & 0 & \phi T & -qT \\ -\omega M_n & 0 & -\eta-\omega C-gM_c & -gM_n & 0 \\ 0 & -eM_c & -dM_c & -dM_n-eT-iD & -iM_c \\ 0 & 0 & 0 & 0 & -\tau \end{pmatrix}$$

Substituting the equilibrium point gives:

$$\begin{pmatrix} -r\gamma-aM_c^* & 0 & 0 & 0 & 0 \\ \lambda & \beta M-\delta+\phi M_c^* & 0 & 0 & 0 \\ -\omega M_n^* & 0 & -(\eta+gM_c^*) & -gM_n^* & 0 \\ 0 & -eM_c^* & -dM_c^* & -dM_n^* & -iM_c^* \\ 0 & 0 & 0 & 0 & -\tau \end{pmatrix}$$

The eigenvalues of this matrix can be found for ξ_1, ξ_2, ξ_3 since the form is in a upper diagonal matrix:

$$\begin{aligned} \xi_1 &= -r\gamma - aM_c^* \\ \xi_2 &= \beta M - \delta + \phi M_c^* \\ \xi_5 &= -\tau \end{aligned}$$

The eigenvalues for ξ_3, ξ_4 can be found by calculating the trace and determinant for the sub-matrix:

$$A_{sub} = \begin{pmatrix} -(\eta + gM_c^*) & -gM_n^* \\ -dM_c^* & -dM_n^* \end{pmatrix}$$

and calculating the characteristic equation:

$$\xi^2 - \mathrm{Tr}(A_{sub})\xi + \det(A_{sub}) = 0$$

Where:

$$\begin{aligned} \mathrm{Tr}(A_{sub}) &= -(\eta + gM_c^* + dM_n^*) \\ \det(A_{sub}) &= (-\eta - gM_c^*)(-dM_n^*) - (-gM_n^*)(-dM_c^*) = d\eta M_n^* \end{aligned}$$

Using the quadratic formula:

$$\xi_{3,4} = \frac{-(\eta + gM_c^* + dM_n^*) \pm \sqrt{(\eta + gM_c^* + dM_n^*)^2 - 4d\eta M_n^*}}{2}$$

Thus the equilibrium points are:

$$
\begin{aligned}
\xi_1 &= -r\gamma - aM_c^* \\
\xi_2 &= \beta M - \delta + \phi M_c^* \\
\xi_3 &= \frac{-(\eta + gM_c^* + dM_n^*) + \sqrt{(\eta + gM_c^* + dM_n^*)^2 - 4d\eta M_n^*}}{2} \\
\xi_4 &= \frac{-(\eta + gM_c^* + dM_n^*) - \sqrt{(\eta + gM_c^* + dM_n^*)^2 - 4d\eta M_n^*}}{2} \\
\xi_5 &= -\tau
\end{aligned}
$$

which all eigenvalues except ξ_2 have negative real part thus creating the conditions for stability. □

Proposition 4. *The cancer, and drug free equilibrium point* $P_0 = (0, \beta M_n^* + \phi M_c^* - \delta, M_n^* = \frac{p}{\eta + gM_c^*}, M_c^* = \frac{h}{dM_n^* + eT^*}, 0)$ *is stable when* $a_1 a_2 > a_3$

with
$a_1 = \eta + gM_c^* + dM_n^* + eT^*$,
$a_2 = e\phi T^* M_c^*(\eta + gM_c^*)$, *and*
$a_3 = \eta d M_n^* + eT^*(\eta + gM_c^*) + e\phi T^* M_c^*$

Proof. Finding the cancer and drug free equilibrium is found by setting $C^*, D^*, u = 0$

$$
\begin{aligned}
\frac{dM_n}{dt} &= p - M_n(\eta + gM_c) = 0 \implies M_n^* = \frac{p}{\eta + gM_c^*} \\
\frac{dM_c}{dt} &= h - M_c(dM_n + eT) = 0 \implies M_c^* = \frac{h}{dM_n^* + eT^*} \\
\frac{dT}{dt} &= T(\beta M_n^* + \phi M_c^* - \delta) = 0 \implies T^* = \beta M_n^* + \phi M_c^* - \delta
\end{aligned}
$$

Using the Jacobian found in Proposition 3, and substituting in the equilibrium point gives:

$$
\begin{pmatrix}
-r\gamma - aM_c^* & 0 & 0 & 0 & 0 \\
\lambda & 0 & 0 & \phi T^* & -qT^* \\
-\omega M_n^* & 0 & -(\eta + gM_c^*) & -gM_n^* & 0 \\
0 & -eM_c^* & -dM_c^* & -(dM_n^* + eT^*) & -iM_c^* \\
0 & 0 & 0 & 0 & -\tau
\end{pmatrix}
$$

Upon inspection, eigenvalues $\xi_1 = -r\gamma - aM_c^*, \xi_5 = -\tau$ can be obtained directly from the matrix as the other column entries are zero. The values will always be negative. The other eigenvalues can be obtained by inspecting the sub matrix and applying Routh-Hurwitz criteria as described in

Appendix B of [6].

$$\begin{pmatrix} 0 & 0 & \phi T^* \\ 0 & -(\eta + gM_c^*) & -gM_n^* \\ -eM_c^* & -dM_c^* & -(dM_n^* + eT^*) \end{pmatrix}$$

resulting in the sub characteristic equation of the form

$$\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3 = 0$$

where $a_1 = \eta + gM_c^* + dM_n^* + eT^*$ is the trace of the sub-matrix, $a_2 = e\phi T^* M_c^*(\eta + gM_c^*)$ is the determinant of the sub-matrix and $a_3 = \eta dM_n^* + eT^*(\eta + gM_c^*) + e\phi T^* M_c^*$ is the sum of the 2x2 principal minors, alongside the fact that all parameters other than β must be positive gives the criteria as needed. □

5 Discussion

Original Model

A critical observation from Section 2 concerns the sign of the cross-dressing parameter β. The original paper treated β only produced bifurcation diagrams and hypersurfaces for positive β yet evidence from DeNardo et al. [3] supports a suppressive role for monocytes in certain tumor micro-environments, thus allowing for negative values of β. Further evidence from the inspiration paper Elewaut et al. indicates that there could be no stimulatory effect indicating $\beta = 0$. Proposition 1 formalities this: whenever $\beta < 0$ and all remaining parameters are strictly positive, the condition $\beta\rho < \delta\eta$ is automatically satisfied and the tumor-free equilibrium E_0 is locally asymptotically stable. These parameter values were not explored numerically in [4].

A second concern is the suggestion in [4] that η may be set to zero. As noted in Section 2, this forces $\rho = 0$ at the tumor-free equilibrium, eliminating monocyte production entirely. Because monocytes are continuously recruited from bone-marrow precursors under homeostatic conditions [7], a model with $\rho = 0$ lacks biological grounding. A solution is to retain $\eta > 0$ as a small but non-zeroclearance rate, which preserves the monocyte source term while still allowing cancer-mediated death ($-\omega CM$) to dominate during active tumor growth.

Macrophage Extension

The extension in Section 3 separates monocytes (M_n) from macrophages (M_c), reflecting monocyte-to-macrophage maturation [7]. The tumor-free equilibrium $P_0 = (0, 0, M_n^*, M_c^*)$ retains a clean stability criterion: the point is locally asymptotically stable when $\beta M_n^* + \phi M_c^* < \delta$. This threshold has an interpretation; stability requires that the combined stimulatory effect of monocytes and macrophages on T-lymphocytes be outweighed by T-cell death. A limitation to this is the model

does not include the suppressive effects on macrophages on T-cells as defined in Reis et al. [7]. Future exploration can include further compartmentalization to describe M1 and M2 macrophages.

Chemotherapy Extension

The chemotherapy extension in Section 4 introduces a standard exponential drug-decay equation and augments the majority of cellular compartments with a drug-kill term. These terms where added under the assumption that chemotherapy attacks cancer and healthy cells alike. The assumption that chemotherapy does not affect monocyte production is a simplification. Work by Larinova et al. [5] summarizes the various effects of chemotherapy agents on macrophages in-vitro. Model expansion to explore the various dynamics is another avenue for further investigation.

Two biologically distinct tumor-free scenarios were analyzed. The first ($T^* = 0$) corresponds to a post-treatment state in which cancer and drug have been cleared but no residual immune activity remains. The stability criterion for this case mirrors that of the macrophage extension, with $\xi_2 = \beta M_n^* + \phi M_c^* - \delta$ again being the sole potentially destabilizing eigenvalue. The second scenario ($T^* > 0$) represents immune-mediated tumor clearance with residual T-cell activity. Stability here is characterized by the Routh-Hurwitz criterion $a_1 a_2 > a_3$ as described in Mathematical Biology by J.D. Murray [6].

6 Conclusion

This paper examined the tumor-immune model of Hernandez-Lopez et al. [4], identifying vague parameter description proposed biologically motivated corrections. Two model extensions were then developed and analysed. The macrophage extension captures the monocyte-to-macrophage differentiation yielding a four-variable system whose tumor-free equilibrium is characterized by the combined immunostimulatory pressure $\beta M_n^* + \phi M_c^*$ relative to lymphocyte death δ. The chemotherapy extension further augments this system with a drug concentration variable and analyses both immune-absent and immune-active tumor-free states, demonstrating that post-treatment immune activity provides an additional layer of stability governed by the Routh-Hurwitz condition $a_1 a_2 > a_3$.

Collectively, these results indicate that model structure and parameter interpretation impact the qualitative dynamics and biological conclusions drawn from tumor-immune systems. In particular, the sign of the monocyte–T-cell interaction parameter β qualitatively changes the nature of the tumor-free equilibrium, and the explicit inclusion of macrophage dynamics reveals stability conditions that are invisible in the reduced monocyte-only model. Future work would include bifurcation analysis using mathematical continuation software and simulation of biological viable parameters from experimental data, which due space and time limitations of this project the proof of concept code was not included but are available at https://github.com/Rosiedefazio/AMATH-881-Final-Project.

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